Sophia Becker

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Education

École Polytechnique Fédérale de Lausanne (EPFL) — Lausanne, Switzerland
PhD in Computational Neuroscience — 11/2020 – 06/2026
- Thesis: *A theory of novelty and exploration in naturalistic environments*, Advisor: Prof. Wulfram Gerstner

Goethe University Frankfurt — Frankfurt, Germany
MSc. Interdisciplinary Neuroscience — 10/2017 – 02/2020
- Completed with distinction (German grade: 1.0, excellent)
- Thesis: *The Role of Slow PRG1 Dynamics in Synaptic Working Memory*, Advisor: Prof. Tatjana Tchumatchenko

Technical University of Darmstadt — Darmstadt, Germany
BSc. Mathematics, bilingual degree — 10/2013 – 05/2017
- Minor in computer science
- Thesis: *Markov Processes on Electrical Networks and Domains*, Advisor: Prof. Frank Aurzada

Hong Kong University of Science and Technology (HKUST) — Hong Kong
Study abroad — 08/2015 – 06/2016

Professional & Research Experience

Doctoral assistent with Prof. Wulfram Gerstner, *EPFL* — Lausanne, Switzerland, 11/2020 – 08/2026
Understanding how *representation-specific generalization* impacts *novelty computation* in the brain and down-stream *novelty-driven behaviors*, using normative modeling, statistical inference, reinforcement learning, mechanistic models of synaptic plasticity, model-based data analysis.

Research Assistant with Prof. Tatjana Tchumatchenko, *MPI for Brain Research* — Frankfurt, Germany, 10/2018 – 08/2020
Understanding how *long astrocytic timescales* impact working memory dynamics in networks with *short-term synaptic plasticity*, using spiking network simulations & mathematical stability analysis.

Research Internship with Prof. Torfi Sigurdsson, *Neurophysiology Institute, Goethe University* — Frankfurt, Germany, 06/2018 – 07/2020
Behavioral training, in-vivo electrophysiology in mice during an auditory working memory protocol.

Research Internship with Prof. Julijana Gjorgjieva, *MPI for Brain Research* — Frankfurt, Germany, 04/2018 – 06/2020
Advanced analysis of Calcium imaging data from mouse visual cortex during development.
in collaboration with Dr. Paloma Maldonado, Netherlands Institute for Neuroscience, Amsterdam.

Research Internship with Prof. Tatjana Tchumatchenko, *MPI for Brain Research* — Frankfurt am Main, Germany, 01/2018 – 04/2018
Partial differential equation model for the spread of neural apoptosis and inflammation in neuro-degenerative diseases.

Natural Science Research Program, *German National Academic Foundation* — Köln / Hannover, Germany, 09/2017 – 04/2019
Neuro-inspired computing based on the dynamics of coupled oscillator networks.
External research project with biannual progress meetings, supervisor: Prof. Marc Timme.

Working student, *IT Innovation department, Merck KGaA* — Darmstadt, Germany, 10/2016 – 10/2017
Machine learning (time series prediction, clustering) for real-time *database* system SAP HANA.

Awards, Fellowships, & Grants

2026 **Trainee Professional Development Award (TPDA)**, Society for Neuroscience
2024 **Teaching Assistant Award**, School of Computer and Communication Sciences, EPFL
2022 **Teaching Assistant Award**, School of Computer and Communication Sciences, EPFL
2014-2020 **Excellence Scholarship**, German National Academic Foundation
2013 **High School Graduate Prize for Mathematics**, German Mathematical Society

Publications

S Becker(2026) A theory of novelty and exploration in naturalistic environments. *PhD Thesis.* EPFL, Lausanne, Switzerland

S Becker*, A Modirshanechi, W Gerstner (2026) Representational similarity modulates neural and behavioral signatures of novelty. *Neuron* 114, 1863-1876.e17

L Gruaz*, A Modirshanechi*, **S Becker**, J Brea (2025) Merits of curiosity: a simulation study. *Open Mind* 9: 1037–1065

S Becker*, A Modirshanechi, W Gerstner (2024) Computational models of intrinsic motivation for curiosity and creativity. Behavioral and Brain Sciences 47, e94

A Modirshanechi*, **S Becker**, J Brea, W Gerstner (2023) Surprise and novelty in the brain. *Current Opinion in Neurobiology* 82, 102758

S Becker*, A Nold*, T Tchumatchenko (2022) Modulation of working memory duration by synaptic and astrocytic mechanisms. *PLoS Computational Biology* 18 (10), e1010543

Conference Talks and Posters

Contributed talks:

- Representational similarity modulates neural and behavioral signatures of novelty.
  Intrinsic Motivation (PIMBAA) workshop, RLDM 2025, Dublin, Ireland.
- Representational similarity modulates neural and behavioral signatures of novelty.
  Cosyne 2024, Lisbon, Portugal.

Selected posters:

- Representational similarity modulates neural and behavioral signatures of novelty.
  RLDM (2025), Foraging Conference (2024), Bernstein (2023)
- A bioplausible implementation of novelty-guided exploration.
  RLDM (2022), Bernstein (2022)
- Transient working memory is controlled by slow synaptic dynamics.
  Cosyne (2020), Bernstein (2019)

Invited Talks

2026 **SfN nanosymposium**, Washington DC, USA
2026 **Gjorgjieva Lab**, Technical University of Munich, Germany (online)
2025 **Haesler Lab**, Katholieke Universiteit Leuven, Belgium
2025 **Swiss Computational Neuroscience Retreat**, Crans Montana, Switzerland
2024 **Moreno-Bote Lab**, University Pompeu Fabra, Barcelona, Spain
2023 **Zero-Noise Lab**, Ernst-Strüngemann Institute, Frankfurt am Main, Germany
2023 **Sensorimotor Symposium**, EPFL, Lausanne, Switzerland

Teaching Experience

since 2023 **Supervision of Master students**, EPFL

- Pingsheng Li (semester projects 2024 & 2025, contributed talk at CCN 2025)
- Max Conti (semester project 2024)
- Yueying Wang (semester project 2023)

2023 – 2024 **Teaching Assistant, Artificial Neural Networks and Reinforcement Learning**, EPFL

- Head of Teaching Assistant Team & Teaching Award 2024
- Development of new course material (1-month course projects, lecture on RL ethics)

2021 – 2022 **Teaching Assistant, Biological Modeling of Neural Networks**, EPFL

- Head of Teaching Assistant Team & Teaching Award 2022
- Development of new course material (1-month course projects)

2021 **Teaching Assistant, Introduction to Scientific Programming in Python**, EPFL

- Development of new course material (automated grading, exercises)

Service, Outreach & Professional Development

Service and Outreach:

2024-2025 **Nature in Code Summer School, EPFL**, Research outreach for Swiss high school students
2022-2025 **Neuro Night Seminar Series, EPFL**, Co-organizer
2023 **EDNE PhD Alumni Event**, Co-organizer and moderator
2020 **German Neuroscience Olympiad**, Co-organizer for the German National Competition
2014 – 2015 **Student Society of the German National Academic Foundation, TU Darmstadt**, Organizer

Courses:

2025 **Summer Workshop on the Dynamic Brain**, Friday Harbor, USA
2025 **Methods in Computational Neuroscience Course**, Marine Biology Labs, USA
2023 **Neurophysics Summer School**, Lausanne, Switzerland
2021 **Computational Psychiatry Course**, Zürich, Switzerland (online)
2016 **Summer Course on Complex Patterns and Self-Organization**, Nice, France

Coding

CODING Python, MATLAB, C^{++}, SQL Script, Java

TOOLS PyTorch, Git, Shell, Kubernetes

Languages

GERMAN native
ENGLISH professional
FRENCH conversational
NORWEGIAN conversational